FABIO SLIKA STELLA

Senior Embedded Linux and Firmware Engineer | C/C++ | Yocto | Kernel Drivers | Bare-Metal

Pato Branco, PR, Brazil | fabioslikastella@gmail.com | github.com/Gedankenn | Portuguese, English

Professional Summary

Computer Engineer shipping firmware and Embedded Linux on production devices (electronic security, railway, automotive). Owns Yocto (BitBake), Device Tree (DTS/DTB), Linux kernel drivers/BSP and bare-metal Embedded C/C++ on ARM Cortex-M (MSP, STM32, ESP32): bootloader, OTA, cryptography, MQTT, Ethernet/TCP-IP, CAN, RS-485, I2C, UART. Works in cross-functional teams; listens closely; treats code review as engineering, not conflict.

Technical Skills

Languages: C, C++, Embedded C, Python, MATLAB, Bash, Assembly
Embedded Linux: Yocto Project, BitBake, OpenEmbedded, Device Tree, DTS, DTB, Linux kernel, kernel drivers, BSP, OpenWrt, GCC
Firmware: bare-metal, bootloader, OTA, board bring-up, ARM Cortex-M, STM32, ESP32, MSP
Connectivity / security: CAN, RS-485, I2C, UART, Ethernet, TCP/IP, MQTT, 4G, cryptography, GDB, Git, CMake, MATLAB/Simulink, Typhoon HIL

Professional Experience

Control iD — Sao Paulo, Brazil

Systems Developer II | 2026 — Present

- Owned production Embedded Linux images (Yocto/BitBake, Device Tree DTS/DTB, Linux kernel drivers) for access-control devices.
- Developed bare-metal Embedded C/C++ on 3 ARM Cortex-M families (MSP, STM32, ESP32): bootloader, OTA and board bring-up.
- Integrated cryptography, Ethernet, TCP/IP, MQTT and web services; led code review and GDB debug on the R&D team.

Progress Rail — Caterpillar

Embedded Systems Engineer | 2024

- Shipped locomotive-controller firmware in C/C++ with CAN, RS-485 and 4G Ethernet via cellular modem (MSP).
- Designed data architecture, persistent storage and microcontroller memory map for field operation.

Zeentech / Stellantis

Software Engineer | 2024

- Built MATLAB/Simulink vehicle models and validated automotive embedded code from simulation to bench.

InoBram

Embedded Systems Developer | 2022 — 2024

- Wrote STM32 bare-metal firmware in C across a 2-year industrial product cycle, including bootloader, OTA and bring-up.

IAV do Brasil

Test Engineer | 2022

- Performed white-box HIL validation of automotive electronics with Typhoon HIL.

Xpert Automacao

R&D Intern | 2021 — 2022

- Implemented ESP32/STM32 bare-metal firmware; deployed OpenWrt on a commercial router adopted as the product network backbone.

Selected Projects

- Built ESP32-FOC — Field-Oriented Control for BLDC motors in C. github.com/Gedankenn/ESP32-FOC
- Delivered Strata — local C++ LLM inference engine (OpenAI/Anthropic API). github.com/Gedankenn/Strata

Education

B.S. Computer Engineering | UTFPR | 2014 — 2022 | FOC research 2020—2022 | Bare-Metal STM32, Udemy 2023

Languages

Portuguese (native), English (advanced)